

Threading dislocation propagation in AlGaIn/GaN based HEMT structures grown on Si (111) by plasma assisted molecular beam epitaxy

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ABSTRACT

A transmission electron microscopy (TEM) study was carried out on a series of AlGaIn/GaN high-electron mobility transistor (HEMT) structures grown by plasma assisted molecular beam epitaxy (PA-MBE) on Si (111). Threading dislocation (TD) behavior and density were investigated for three heterostructures using an AlN/GaN superlattice and/or differently strained GaN layers. Threading dislocation densities (TDDs) were measured by TEM (at different depths) and high resolution x-ray diffraction (HRXRD) allowing one of the most complete and few studies so far, presenting separated values on edge, screw and mixed type TDs quantities.

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1. Introduction

High power devices have evolved during the last thirty years from structures based on III-As to III-N semiconductors, due to both improvement in group-III nitrides growth processes [1] and the fact that III-N based devices proved to be able to manage up to ten times more power density than their III-As analogues [2]. Among these high power devices, AlGaIn/GaN high electron mobility transistors (HEMT) have been reaching a more and more considerable stage of development and commercialization during the last two decades [3].

However, unlike in the case of arsenides, III-N compounds present an important bottleneck for their higher industrial expansion: at the present moment, good-quality large enough group-III nitride wafers or bulk crystals are still not available. For this reason, devices implemented with these materials are grown on non-native substrates. Without a doubt, α -Al₂O₃ (0001) (sapphire) has been the most extensively used, since the demonstration of candle class LEDs [4]. Nevertheless, other materials offer better results acting as a substrate which is explained by two reasons involving both top layer and substrate: (i) the lattice parameter mismatch between III-N compound and substrate; and (ii) the substrate thermal expansion and conductivity. The last mentioned issue is extremely important, as self-heating is the

main problem encountered in high power devices (and therefore in AlGaIn/GaN HEMTs) [5] and therefore the power density (which implies temperature increment) in the device is dissipated precisely through the substrate [6].

Lahrèche et al. elaborated a complete summary of the different substrates (such as SiC, Si, ZnO, LiAlO₂, etc.), and their relevant properties, used for growing hexagonal (wurtzite) GaN and AlN [7]. Attending to some physical properties, 4H-SiC (0001) substrates lead to the best electronic responses [8], since it presents a $\sim 3.5\%$ ($\sim 1\%$) mismatch with GaN (AlN), and the highest thermal conductivity (3.8 W/cm). Sapphire, on the contrary, has a poor thermal behavior (with a thermal conductivity about six times lower than SiC), this being the reason for the small thermal resistance occurring due to some malfunctions in HEMTs on sapphire, such as the fact that drain current does not reduce with the increase of the source–drain voltage [2]. Despite these reasons, high cost of SiC constitutes a clear disadvantage when considering it as a practical substrate in an industrial context.

Hence, Si (111) substrates represent a very attractive alternative; since it partially solves the power dissipation problem (it has a thermal conductivity value of 1.5 W/cm). Moreover, the possibility of using high quality, low cost, largely available Si and its well established integrated microelectronic technology compensates widely the inferior but improvable electronic results obtained compared to the ones that are achieved with SiC; actually, Si is nowadays the most reasonable choice for industrial purposes.

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On the other hand, the mismatch between lattice parameters of substrate and nitride semiconductor ($\sim 17\%$ for the GaN/Si (111) system) produces defects in the epilayer in order to relax the structural stress, such as stacking faults, inversion domains, and misfit and threading dislocations (TD) [9], the latter being the most common among them [10–12]. Also, it has been concluded that TDs in these semiconductors behave as current leakage paths [13] and as non-radiative recombination centers [14,15], limiting electronic devices' efficiency and lifetime (and even in the case of opto-electronic devices; although for III-N LEDs and lasers, TDs do not have such a critical effect as for their III-As equivalents) [10]. These dislocations are even the origin of some surface defects, such as pits [16]. Therefore, the study of their behavior, as well as the quantification and reduction of their density, becomes a fundamental topic through the engineering of III-N heterostructures.

A certain lattice mismatch is currently unavoidable but a reduction of TDs is required. Efforts in this line have been focused on (i) researching modifications of parameters and/or steps in the growth process, and (ii) using additional buffer-like layers to induce the recombination of these defects. For AlGaIn/GaN systems, one evident approach for reducing the dislocation density consists in increasing the GaN layer thickness [17] (this is also valid in most systems) [18], but generally better results have been achieved using low temperature AlN layers, AlN/GaN superlattices (SL) [19], $\text{Al}_x\text{Ga}_{1-x}\text{N}$ where the composition x varies along the growth direction [20,21], or other interesting solutions, such as using CrN layers [22], Si δ -doping [23] or even He implantation [24].

Although molecular beam epitaxy (MBE) played a secondary role in the production of GaN based HEMTs on SiC substrate, the technology might have significant advantages at the growth of bow free GaN devices on Si substrate [25]. Recently, plasma assisted MBE (PAMBE) proved to be suitable for the production of AlGaIn/GaN HEMT devices on 4" SiC substrate with excellent uniformity and reproducibility [26]. One fundamental difference between MOVPE and MBE growth of GaN based structures on silicon substrate is the considerably lower growth temperature, which is in case of PAMBE about 700°C in contrast to approximately $1000\text{--}1100^\circ\text{C}$ at MOVPE growth. The thermal induced tensile stress caused by the different thermal expansion coefficients of silicon and GaN is only about 0.2% for MBE growth instead of ca. 0.3% for MOCVD growth. Hence also the compressive strain during GaN growth needs to be lower, which is easier to control and reduces the risk of damage of the grown films during the cooling down process.

In this work, we studied the influence of the strain state of the epitaxial layer on the defect density in the grown layers on silicon substrate in order to identify the growth process resulting in the heterostructure with the lowest TDD.

2. Displaying threading dislocations

Structural defects characterization methods are an essential tool to lower their population in the materials and optimize growth procedures. As it is well known, in the case of TDs, transmission electron microscopy (TEM) applied to planar view-TEM preparations leads to the most reliable results [27], although other techniques, such as cathodoluminescence, atomic force microscopy, or X-ray diffraction (XRD) also provide valuable information.

In the present work, TDs in several samples have been studied by TEM. In this way, the characterization helped to select and adjust the growth parameters for GaN-on-Si state-of-art HEMTs. These structures and its growth processes are wider described in

previous works [28]. Some TEM results on TD quantifications in similar materials have been presented [16,21]; however, the study presented in this work is one of the most extensive ones for TDs in wurtzite materials using the combination of planar view-TEM (PVTEM) and cross section-TEM (XTEM) preparations of the observed samples and recent understanding on the TD observation [29]. Both approaches to the problem permit a highly reliable measurement of TD densities (TDDs), and therefore separated data on three different types of TDs according to their Burgers vector (screw, edge and mixed).

To obtain images of highlighted TDs, samples must be oriented following the invisibility criteria for the distortion associated to a network defect, in the present case, a contrast associated to the lack of dispersion if a TD with the Burgers vector $b^\rightarrow$, since $g^\rightarrow \times b^\rightarrow = 0$, where $g^\rightarrow$ is the reciprocal space vector of the excited reflection. Attending to their characteristic $b^\rightarrow$, TDs are classified into edge, screw and mixed, which, in an hexagonal system, correspond to $b^\rightarrow_a = \frac{1}{3}\langle 11\bar{2}0 \rangle$, $b^\rightarrow_c = \langle 0001 \rangle$ and $b^\rightarrow_{a+c} = \frac{1}{3}\langle 11\bar{2}3 \rangle$, respectively. In this way, obtaining TEM images that contain only one type of TD (screw, edge or mixed) is geometrically not possible. Traditionally, this led to the fact that TDD could only be calculated in groups ("edge+mixed" and "screw+mixed"), but not separately.

Currently, this handicap might be overcome. Based on a method proposed by Follstaedt et al., it has been proved that using the reflection $g^\rightarrow = (2\bar{1}\bar{1}1)$ near the $[1\bar{2}1\bar{3}]$ zone axis, which is 31.4° away from the $[0001]$ direction, it is possible to reveal all types of TDs clearly inside a single TEM diffraction contrast image [29].

Table 1 shows the zone axis and related reflections selected to acquire those TEM images through which TDDs measurements were carried out in this work, as well as the types of TDs that could be observed using each reflection.

The possibility of quantifying all TDs, together with the values for edge+mixed TD and for screw+mixed TD, allows determining the values of edge, screw and mixed TDDs in a separated way. This can be done by a simple equation system, presented in Eqs. (1)–(3), where the indexes e , s and m stand for each type of threading dislocation and other indexes indicate the density obtained through the given reflection, according to Table 1

$$D_s = D_{(2\bar{1}\bar{1}1)} - D_{(11\bar{2}0)} \quad (1)$$

$$D_e = D_{(2\bar{1}\bar{1}1)} - D_{(0002)} \quad (2)$$

$$D_m = D_{(11\bar{2}0)} + D_{(0002)} - D_{(2\bar{1}\bar{1}1)} \quad (3)$$

Note that to achieve a trustable measurement of TDDs, an extra adjustment has to be done: the value obtained through the reflection $(01\bar{1}0)$, as shown in Table 1, is equaled to the analogous edge+mixed density calculated through PVTEM (since the density values for this preparation are much more reliable) [27]. This allows obtaining the thickness (t) estimated for the projection of material in the XTEM preparation along the electron beam path or the TEM optical axis (which, otherwise, should be a priori

Table 1
Type of threading dislocations revealed by each TEM preparation, zone axis, reflections.

Zone axis	Reflection	TD
PVTEM		
[0001]	(11 $\bar{2}$ 0)	Edge and mixed
[1 $\bar{2}$ 1 $\bar{3}$]	(2 $\bar{1}$ 10)	Edge, screw and mixed
XTEM		
[11 $\bar{2}$ 0]	(0002)	Screw and mixed
[11 $\bar{2}$ 0]	(01 $\bar{1}$ 0)	Edge and mixed

calculated through simulations or thickness maps), according to Eq. (4), in which n stands for the number of TDs crossing the length l in the XTEM image which is perpendicular to both the growth direction and the TEM optical axis. This adjustment is made using TDD values near the surface in XTEM preparation (and not at the HEMT/substrate interface), due to the fact that PVTEM images only collect TDs near the surface (electron-transparency is not possible far below 200 nm thickness, depending on the material) and TDD decreases in the material with the proximity to the surface (due to TD annihilations, etc.). However, the calculated t thickness is still valid when calculating the TDD values at the HEMT/substrate interface through XTEM images and using

$$D_{\text{XTEM}} = \frac{n}{lt} \quad (4)$$

3. Experimental methods

All samples were grown in a Veeco Gen20A system capable in forming wafers of diameters up to 4". The chamber is equipped with conventional effusion cells for Ga and Al with high capacitance. Active nitrogen was provided by two Veeco Unibulb nitrogen plasma sources each operated at 400 W and a N₂ gas flow of 4 sccm, yielding a nitrogen limited growth rate of 0.86 $\mu\text{m/h}$. For charge carrier compensation in the GaN buffer layer, it was doped with carbon using CBr₄, which was introduced through a leak valve into the growth chamber and cracked at the hot substrate surface. A LAYTEC EpiCurve curvature measurement system based on two parallel laser beams in combination with an EpiTT reflectivity measurement system and reflectivity corrected pyrometer was used to measure wafer bow during growth.

The strain at a certain thickness can be calculated using the derivation of the Stoney Formula [30] with a correction from the non-linear theory [31], which leads to Eq. (5), where κ stands for the relative change of curvature; $M_{\text{III-N}}$ and M_{Si} for the biaxial module; $h_{\text{III-N}}$ and h_{Si} for the thicknesses of III-N layer sequence and the Si substrate, respectively; and c is a correction coefficient ($c=0.7$). For the calculations, $M_{\text{III-N}}/M_{\text{Si}}=1.7$ and $h_{\text{Si}}=675 \mu\text{m}$ has been used. Details of the analysis are described elsewhere [28] and are not repeated here

$$\varepsilon(h) = \frac{d\kappa}{dh_{\text{III-N}}} \left(6c \frac{M_{\text{III-N}}}{M_{\text{Si}}} \frac{1}{h_{\text{Si}}^2} \right)^{-1} \quad (5)$$

The layer sequence of the three samples investigated in this study consists of an AlN nucleation layer, an optional AlN/GaN superlattice (Sample S3), a thick GaN buffer layer, a 22 nm thick Al_{0.22}Ga_{0.78}N barrier and a 3 nm thin GaN Cap layer on 4" Si (111) substrate. Samples S1 and S2 have nominal identical structures, but – by choosing different growth conditions of the GaN buffer layer – they have different strains. In S3 an AlN/GaN superlattice (SL) was inserted between AlN nucleation layer and GaN layer. Details of the growth process are given elsewhere [28]. The investigated structures are given in Table 2.

Characterization of the samples using HRXRD was performed with a Panalytical MRD system equipped with a $2 \times \text{Ge } 220$ monochromator (Cu-K α_1 -radiation) and a $3 \times \text{Ge } 220$ analyzer. High resolution analyses were made in order to determinate for the different layers—their lattice parameters, compositions, layer thicknesses and mosaic structure.

To characterize the heterostructures through TEM, samples were thinned to electron-transparency both for XTEM and PVTEM preparations. Conventional TEM (C-TEM) and bright (DC-BF) and dark field (DC-DF) micrographs in diffraction contrast and two-

Table 2

Structure of investigated samples.

	S1	S2	S3
Substrate	Si (111)	Si (111)	Si (111)
AlN-nucleation layer (nm)	130	130	130
SL			10 nm $\times$ 2 nm AlN/18 nm GaN
GaN:C (μm)	1.75	1.75	0.75
GaN (μm)	0.25	0.25	0.25
Al_{0.22}Ga_{0.78}N (nm)	22	22	22
GaN (nm)	3	3	3

beam conditions were taken with a JEOL-1200EX microscope. TDDs (near both the surface and the HEMT/substrate interface) for each sample were calculated as the mean values for about ten different areas ranging between 0.5 and 1 μm^2 (PVTEM) or for about ten different lengths between 1 and 3 μm (XTEM).

The errors for the TD final densities have been calculated through error quadratic propagation, considering a 5% relative error when computing the number of TDs, and a maximum error of 2.5% in the value of the area including these TDs.

4. Results

Bow measurements during growth of the samples allowed the observation of bow changes caused by increasing layer thickness and therefore an analysis of the strain at different depths of the GaN layers as shown in Fig. 1 could be performed. In both Samples S1 and S2, GaN growth started with a compressive strain greater 2% caused by its 2.5% larger in-plane lattice parameter than AlN. Then the strain decreased continuously, changed sign around 400 nm and 700 nm thicknesses for Samples S1 and S2, respectively, and leveled off at $\sim +0.16\%$ for both samples. The mean strain of S2 is almost 0, or in other words, the wafer has a very low bow. In S1, the positive strain of the upper part does not compensate the negative strain of the lower part. The mean strain remains negative and the wafer has a negative bow.

HRXRD θ - 2θ and ω scans (Fig. 2), as well as Reciprocal Space Maps (RSM, not shown here) were acquired to obtain the Al_xGa_{1-x}N layer composition and the lattice parameters for all layers. These results are presented in Table 3, in which strain values have been calculated using Eq. (6), where d stands for each one of the lattice parameters and the index "0" indicates the parameter when the layer is relaxed, which, for the GaN, have been taken as $a_0=3.1884 \text{ \AA}$ and $c_0=5.1850 \text{ \AA}$ [32]

$$\varepsilon_d = \frac{d-d_0}{d_0} \quad (6)$$

Selected area electron diffraction (SAED, not shown here) confirmed that all samples were single-crystalline and they were oriented in the directions $[11\bar{2}0]\text{III-N}/[\bar{1}10]\text{Si}$.

DC-BF images in 2-Beam condition (as well as DC-DF images, which are not shown here) were taken using the criteria and reflections indicated in previous sections for all samples. Figs. 3 and 4 show some areas for PVTEM preparations of the samples in which edge plus mixed TDs and all three types of TDs, respectively, are visible. XTEM images for reflections (01 $\bar{1}0$) and (0002) are presented in Figs. 5 and 6, respectively.

Note that, when possible, images in Figs. 3–6 have been presented using the same scales, which allow comparing visually TDDs near the surface (for PVTEM images) and also at different depths (for XTEM images).

TDD values are presented in Tables 4 and 5. Densities have been measured at three different distances: h_1 and h_2 , which correspond to $\sim 50 \text{ nm}$ and $1230 \mu\text{m}$, respectively, from the AlN

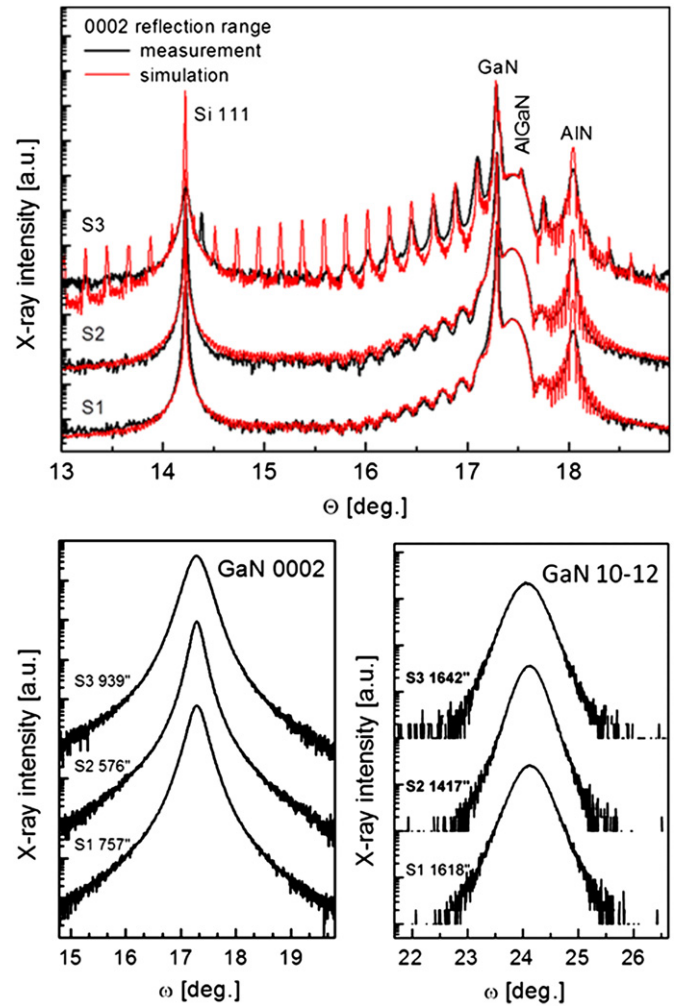
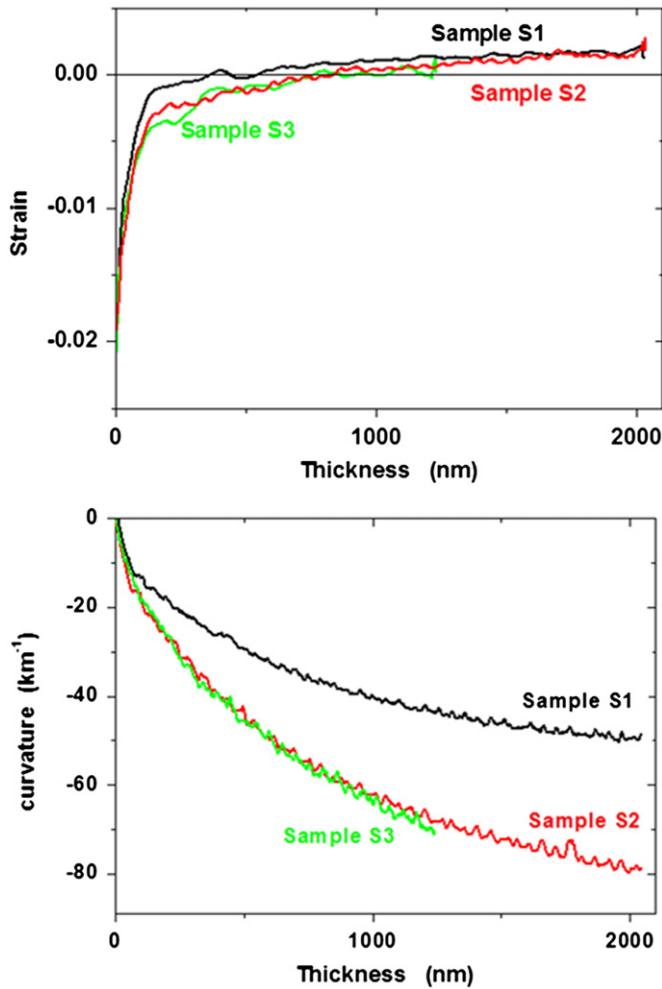


Fig. 1. Calculated depth profiles of strain and curvature at room temperature within GaN buffer layers for all samples.

Fig. 2. θ - 2θ and ω scans for all samples.

nucleation layer (see Table 4); and h_3 , which indicates a distance between 5 and 15 nm below the AlGa barrier layer (see Table 5), except in the case of S2, for which this distance is 175 nm. Note that, only for S3, distances h_2 and h_3 coincide.

Due to the fact that results of TDD measurements in PVTEM preparations are valid only near the surface, as previously commented, mixed-type TDDs are not possible to separate those for edge or screw ones, and hence density values are not presented independently in Table 4. Although these densities have not been calculated through PVTEM images, the XTEM preparations thicknesses (t) have been adjusted to PVTEM results, increasing, in this way, the reliability of the TDDs in this table.

Relative errors in Table 5 are higher than in the case of Table 3 due to the error propagation through Eqs. (1)–(3). The density error for one type of TD influences those for the other types, and, in consequence, some of the lowest densities (as for S1 mixed TDs) present errors that are even higher than the mean value.

5. Discussion

The crystallographic orientation for all heterostructures, observed through SAED, is the one expected when growing hexagonal nitrides on cubic Si, since the threefold symmetry of the Si (111) surface and the six-fold atom arrangement for Si when using this particular growth direction leads to an

Table 3
 ω -scans FWHM and lattice parameters obtained by HRXRD for GaN layer on Si (111). GaN strains are also presented.

Sample	S1	S2	S3
FWHM ω -002 (deg.)	0.2102	0.1601	0.2703
FWHM ω -102 (deg.)	0.4495	0.3764	0.4562
a (Å)	3.1932	3.1926	3.1875
c (Å)	5.1827	5.1831	5.1866
ε_a (%)	0.15	0.13	−0.03
ε_c (%)	−0.04	−0.04	0.03

orientation of $[11\bar{2}0]_{\text{III-N}}/[1\bar{1}0]_{\text{Si}}$. This is why GaN is more easily grown on Si (111) than on Si (001) [33].

Strain depth profile in Fig. 1 shows how strain in GaN is different for all three samples, although Samples S1 and S2 present similar values when approaching the surface. This is also confirmed by XRD data in Table 3 (it is important to understand that XRD is mostly a surface technique), in which it can be observed that the a -axis parameter (and, therefore, strain in the growth plane) presents similar values for Samples S1 and S2. The broadening in FWHM HRXRD curves of Samples S1 and S2 indicates that, near the surface, S2 presents a lower TDD value than S1, which disagrees with the TDD values in Table 5. No reasons have been found for this difference in the results, but it should be taken into account that, although TEM results are statistically significant (since values in Table 5 were calculated

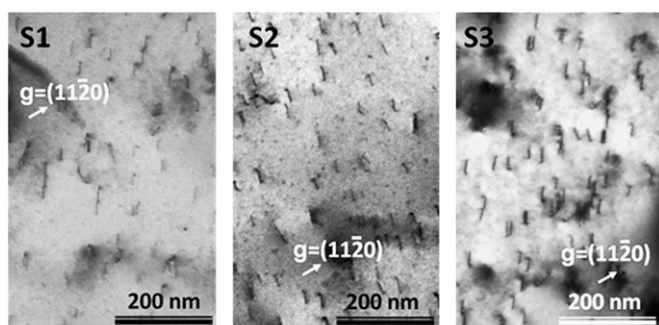


Fig. 3. Edge and mixed TDs in PVTEM preparations for all samples.

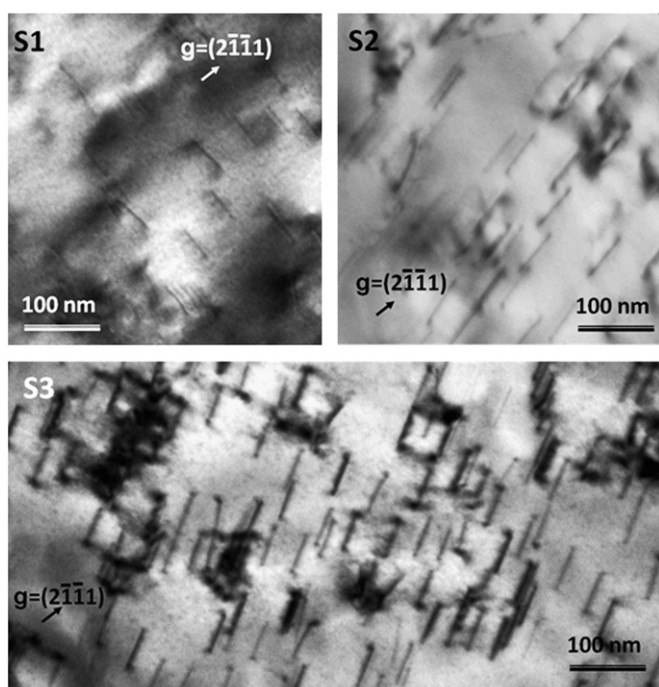


Fig. 4. Edge, screw and mixed TDs visible in PVTEM preparation for all samples.

using the TDDs in at least ten different areas for each sample), HRXRD measurements cover larger areas of the sample.

The goal of these samples TEM characterization is to identify the most convenient growth process on silicon substrate according to the TDD values. Taking into account that this density decreases with the GaN thickness (which is different in all samples), measurements from Sample S3 must be compared to those of the other samples at the same distance over the substrate (h_2), not just at the surface of all heterostructures. Therefore, TDDs in Table 4 (taking into account both edge+mixed and screw+mixed data) indicate that, at a distance h_2 , the heterostructure using an SL obtains the lowest TDD values. This reduction of TDD when an AlN/GaN SL is used is expected, since, as previously commented, SLs are well known ways to favor TD recombination. It should be noticed that the other works in the literature mostly use PVTEM results (as they are the most reliable ones) to determinate these densities, and, consequently, none or very few TEM results have been presented until now on the TDDs formed at the interface between thick GaN layers and Si substrate. TDD measurements at different positions than the surface can only be reliably done through the combination of results for XTEM and PVTEM preparations (as it has been done for h_1 and h_2 in this work).

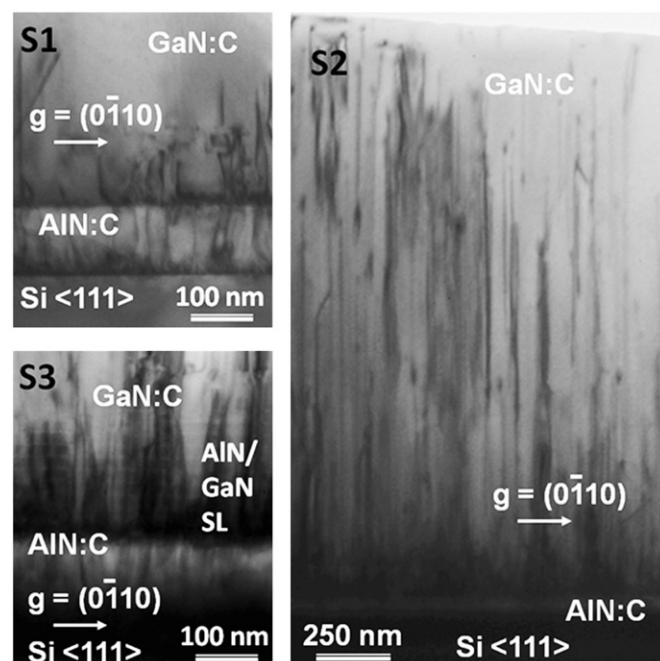


Fig. 5. Detailed and general views for XTEM preparations of the samples using the reflection (0110).

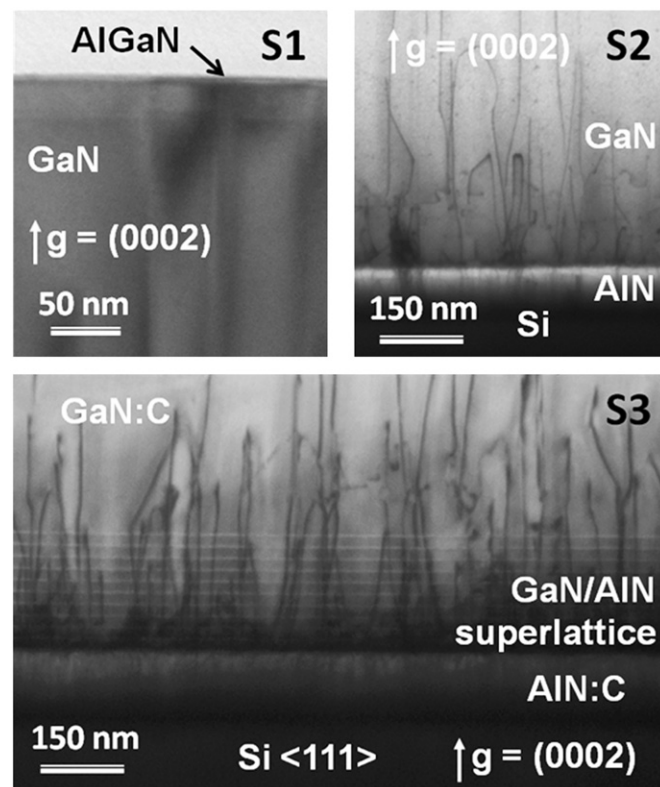


Fig. 6. AlGaIn/GaN and GaN/(SL)/AlN/Si details and general view for XTEM preparations and (0002) reflection of all samples.

Nevertheless, from the device point of view, it is important to analyze the amount of TDs that reach or cross the AlGaIn barrier layer, since those are the ones that most deteriorate the HEMT electronic behavior. Density values near the surface are presented in Table 5 (a general view of the TDs reaching the surface can be seen in Figs. 3 and 4). Densities in this table range from 10^9 to

Table 4
Threading dislocations densities at distances h_1 and h_2 .

Sample	Distance	TDD (10^9 cm^{-2})	
		Edge + mixed	Screw + mixed
S1	h_1	24.4 ± 1.5	42 ± 3
	h_2	15.9 ± 1.0	33.8 ± 2.1
S2	h_1	133 ± 8	76 ± 5
	h_2	52 ± 3	23.7 ± 1.4
S3	h_1	35.9 ± 2.2	22.8 ± 1.4
	h_2	22.8 ± 1.4	16.4 ± 1.0

Table 5
Threading dislocations densities at distance h_3 .

Sample	TDD (10^9 cm^{-2})		
	Edge	Screw	Mixed
S1	6.4 ± 2.0	11.0 ± 1.7	1.9 ± 2.5
S2	11.5 ± 2.5	4 ± 3	11 ± 4
S3	15 ± 3	9 ± 3	7 ± 4

Table 6
Annihilation percentages along the heterostructures referenced to the TDs at distance h_1 .

Sample	Distance	Annihilated TDs (%)	
		Edge + mixed	Screw + mixed
S1	h_2	35	20
	h_3	66	69
S2	h_2	61	70
	h_3	83	80
S3	h_2 (idem for h_3)	36	28

10^{10} cm^{-2} , which is in accordance to, or even improves, results about threading dislocations in GaN on Si found in the literature [21,34]. Although S1 presents lower TDD than the other samples, values in Table 5 are too similar to deduce the best growth process based only on this criterion.

Studying how the different heterostructures promote TD annihilation processes is also a good way to compare the samples. Table 6 shows the percentage of TDs annihilated at distances h_2 and h_3 referenced to that at distance h_1 . The TDD values reduction adjusts reasonably well to an exponential for most cases, as it could be expected [35].

It can be observed that Sample S2 growth process favors the TD annihilation at a much higher rate than those of the other samples. The strain compensation in S2 heterostructure (Fig. 1) clearly explains this good structural behavior, which makes S2 growth conditions technologically more favorable from the point of view of (i) the wafer curvature, since higher bow of the wafer for S1 and S3 could result in process errors; (ii) layer strain state, taking into account that this strain could produce, besides structural defects, unexpected electronic behaviors; and (iii) the structural defects, since (as the FWHM in XRD curves show) S2 presents the lowest density of defects near the surface, and (as TEM characterization demonstrates) the highest TD annihilation and a TDD near the surface close to the lowest (which corresponds to S1).

When comparing results in Table 5 for S3 at distances h_1 and h_2 , it is concluded that AlN/GaN SL is not such an effective filter for threading dislocations as expected a priori. Fig. 6 exposes this information more evidently. DC-Bright Field images of S3 show

how TDs clearly remain mostly unfiltered through the AlN/GaN SL. Since these superlattices are grown precisely as a TD-filter for GaN-based devices on sapphire [36], this result could be shocking, although other authors have found similar behaviors for the use of AlN interlayers in GaN-on-Si systems [21]. It has to be considered that edge and screw TDs behave differently with respect to the annihilation process, and that AlN/GaN superlattice structures filter screw-type TDs than edge-type ones in a more effective way. Therefore, TDs remaining unfiltered is explained because only $\sim 29\%$ of TD is screw-type in S3 (as shown in Table 5). This percentage would have been screened by mixed TDs if the TEM study would not include both XTEM preparations and non-conventional reflections.

It is possible to estimate by semi-empirical methods a ratio between edge, screw and mixed TDDs [37]. This has been widely proven for GaN on SiC, in which edge dislocations represent 90% of all TDs [38]. Nevertheless, for GaN on Si (111) there are no similar semi-empirical predictions. Since the lattice mismatch is similar, it is reasonable to assume that GaN on sapphire would present a similar edge/screw ratio than that of GaN on Si. Therefore, for the studied samples, $\geq 50\%$ of TD would be expected to be pure-edge [39]. Results for Samples S2 and S3 (within their respective error margins) are in good agreement with this. It is of interest to notice that in these cases mixed TDs cannot be neglected at all, since they represent up to 40% of the total TDs. This information corroborates the importance of a separate quantification for TDDs, such as the one presented in Table 5. When these densities are shown grouped, as it is done in Table 4, mixed-type TDDs screen the real ratios between the different types of dislocations.

The case of S1 is the only one that presents an unexpected result at this point, since screw TDs are almost twice numerous than edge ones, which is not common for GaN grown on Si using an AlN buffer layer, but it has been observed when using a ScN buffer layer instead [40].

6. Conclusion

A series of GaN-based HEMT on Si (111) substrate has been grown under different conditions and with different structures and characterized by TEM and HRXRD. A complete and precise TEM method to determinate TDDs (at different distances to the substrate and separately for edge, screw and mixed types) has been presented and applied to the samples, concluding that the use of a GaN thick layer that fully compensates the thermal and structural strain (heterostructure in Sample S2) improves not only the wafer bow, but also the TD annihilation and leads to one of the lowest densities of TDs reaching the barrier layer. The separated values for edge, screw, and mixed TDs obtained with this method had allowed explaining these defects' propagation in the samples, which is not possible by standard approach to determinate TDDs (screened by mixed TDs).

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